Theme B: Air Pollution Project Report (and guidelines)

Total number of points: 13

1) For each of the 15 listed countries (Table 2 in the paper) is there any significant trend (at 5% confidence level) in the outdoor air pollution (timeseries of annual PM10) in the given period (1970-2008)? How do results change if you use annual PM10 per capita instead? [Note: for the latter question, you need to divide the total PM10 (Gg) with the population number for the overlapping time period.]

Instructions:

Fill in the table below with your results, ensuring that you use the correct units for the trends (Gg/yr). Choose a significance level (e.g., 5%) and maintain it consistently for all countries. Note that PM10 values are available for each year, while population data is provided every fifth year. Determine the best method to interpolate the population numbers for the intervening years to obtain annual data. Optional: You may include notes or a figure in the space provided below the table; however, please limit it to one figure. If you include a figure, ensure it has a caption and that all axes are labeled, along with any necessary legends.

Answers:

Significance level used in the results below: __5___%

Country	Trend of PM10 (Gg/yr)	Significant trend? (Y or N)	Trend of PM10 per capita (Gg/yr)	Significant trend? (Y or N)
China	207.1187834	Y	0.074073321	Y
India	206.8267024	Y	0.067616015	Y
Pakistan	9.449828138	Y	-0.006071551	N
Bangladesh	3.786230567	Y	-0.065721514	Y
Nigeria	36.28959757	Y	0.04379588	N
Russia	-1.888410729	N	-0.094026513	N
USA	-197.6871822	Y	-0.975000048	Y
Indonesia	122.7242551	Y	0.199080551	N
Ukraine	-17.36040992	Y	-0.353148573	Y
Vietnam	20.02712854	Y	0.059780807	Y
Egypt	3.064024899	Y	-0.014106998	Y
Germany	-54.75908117	Y	-0.704406074	Y
Turkey	2.808919433	Y	-0.09415128	Y
Iran	1.886200506	Y	0.009910211	Y
Japan	-19.39365992	Y	-0.197866889	Y

Point scheme: 3/15 point for each row above (3 points in total)

Notes (optional):

2) According to the results from the above question, which country out of those 15 has the highest trend in PM10 (Gg) over the last 10 years (1999-2008)? Take this country as a reference and find how correlated its PM10 timeseries is to each of the

PM10 timeseries from other 14 countries, for 1970-2008. Are the correlations significantly different from zero at the 5% confidence level?

Instructions:

Rank the countries below based on their recent trends, from the highest magnitude (rank 1) to the lowest (rank 10). Identify the reference country and provide the correlation coefficients (r) in the table, indicating whether the correlation is significant. To assess significance, calculate the p-value for each correlation. You may include notes in the space provided below the table, but this is optional.

Answers:

Country	PM10 trend for 1999-2008 (Gg/yr)	Rank	Correlation (r) with the reference country	Significant correlation? (Yes or No)
China	328.0587879	3	-0.0514	No
India	199.066485	5	-0.04068	No
Pakistan	2.82893333	11	0.053295	No
Bangladesh	-2.3840061	13	0.022933	No
Nigeria	269.761691	4	-0.34347	Yes
Russia	-573.73354	1	1	N/A
USA	-152.06012	6	-0.04691	No
Indonesia	543.855636	2	-0.013147	No
Ukraine	-1.6906303	14	-0.09711	No
Vietnam	15.4316364	8	0.193648	No
Egypt	2.74488485	12	0.019979	No
Germany	-2.9597394	10	-0.07482	No
Turkey	6.61313333	9	-0.03957	No
Iran	0.17059091	15	0.02319	No
Japan	-39.444697	7	0.052686	No

Point scheme: 1 point for each column above (4 points in total)

Notes (optional):

3) Consider the premature mortality rate per country (=deaths per capita, i.e. number of deaths divided by total population per country) as given Table 2 of Lelieveld et al. (2015). Is there a linear relationship (and how strong is it) between the premature mortality rate per country and the total PM2.5 concentration per country, across those 15 countries? [Note: for each country calculate its premature mortality rate for 2010 (Table 2; 'Deaths' column) and use the total PM2.5 values per country for 2014]. Is there a linear relationship (and how strong is it) between the premature mortality rate per country and urban PM2.5 concentrations, across those 15 countries.

Instructions:

Fill in the tables below with your answers. In the second table, if you determine that the relationship is best represented by a linear equation (Y = a*X + b), write down the equation with your calculated values for a and b. Include the correlation estimate (r) between Y and X and indicate whether the correlation is significantly different from zero at a 5% significance level. Additionally, create two figures:

Figure (A): A scatter plot showing the premature mortality rate per country (y-axis) versus total PM2.5 value per country (x-axis).

Figure (B): A scatter plot showing the premature mortality rate per country (y-axis) versus urban PM2.5 per country (x-axis).

For both figures, add a caption, ensure all axes are labeled, and include any legends as needed. You may include notes in the space provided below the figures, but this is optional.

Answers:

Country	Premature mortality rate for 2010 (death per capita)
China	0.001011955
India	0.000523971
Pakistan	0.000652772
Bangladesh	0.000606792
Nigeria	0.000558256
Russia	0.000468014
USA	0.00017749
Indonesia	0.00021522
Ukraine	0.001117269
Vietnam	0.000497974
Egypt	0.000426616
Germany	0.000422702
Turkey	0.000442539

Iran	0.000350154
Japan	0.000196356

Point scheme: 1 point for the column above

Premature mortality versus:	Relationship (equation)	Correlation coefficient (r)	Significant correlation? (Yes or No)
Total PM2.5	y=2.5490*10^(-6)X+0.000413970	0.2592	No
Urban PM2.5	y=2.4021*10^(-6)X+0.000413535	0.2637	No

Point scheme: 1 point for each row above (2 points in total)

Figure A:

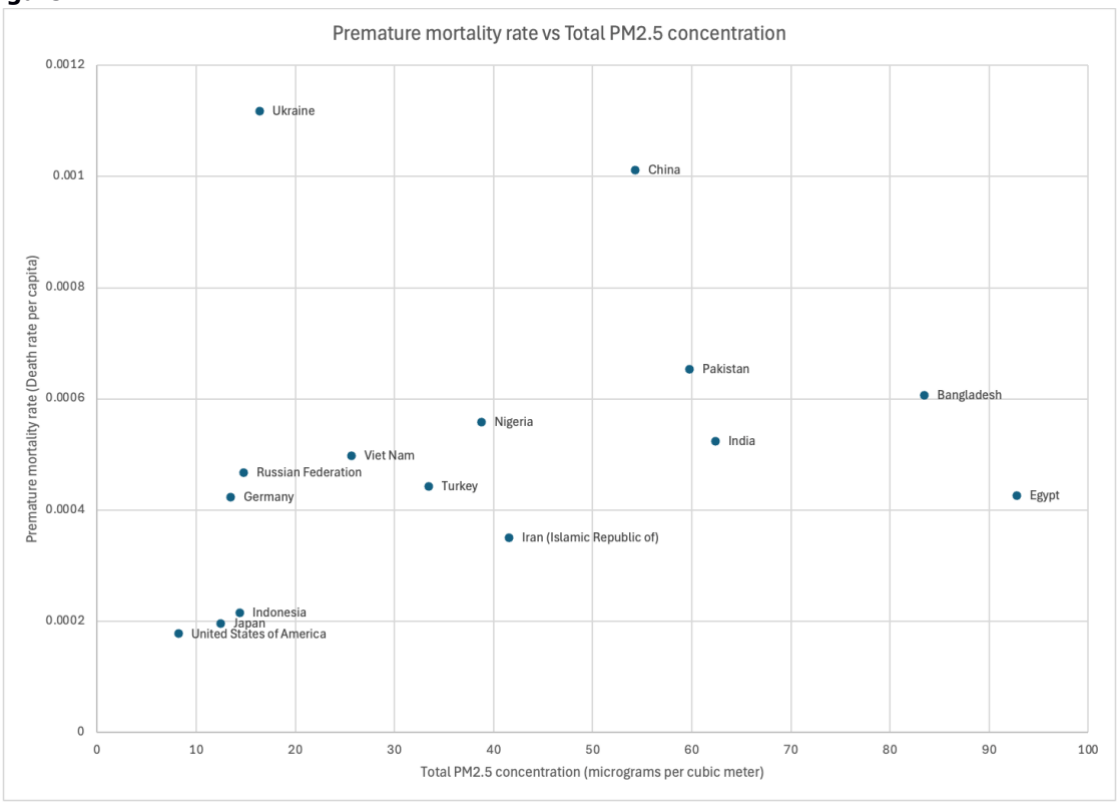


Figure A: Shows the relationship of total PM2.5 concentration (micrograms per cubic meter) vs premature mortality rates (death rate per capita). Each point on the figure represents a country with its name on the right.

Figure B:

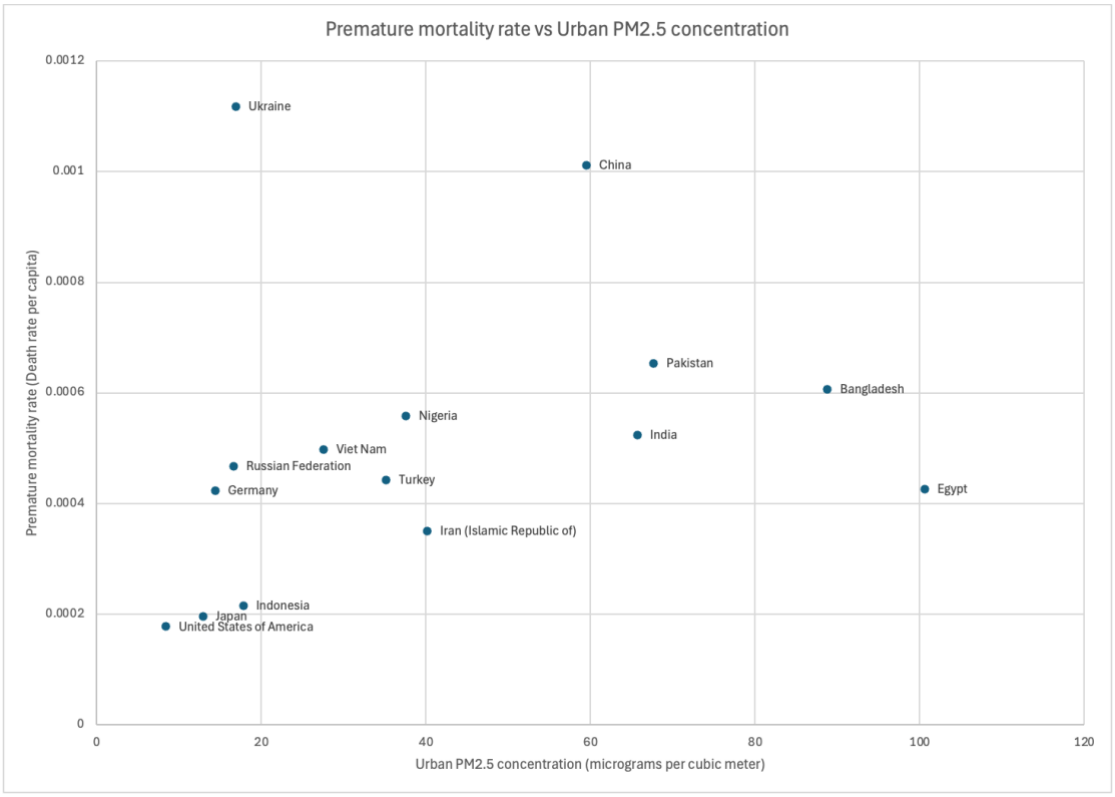


Figure B: Shows the relationship of Urban PM2.5 concentration (micrograms per cubic meter) vs premature mortality rates (death rate per capita). Each point on the figure represents a country with its name on the right.

Point scheme: 1 point for each figure (2 points in total)

Notes (optional):

What is the relationship between healthcare spending as a percentage of GDP and premature mortality rates from outdoor air pollution in the fifteen selected countries in 2010?

Your OWN Proposed Additional Analysis Results

Country	Premature mortality rates 2010 (death per capita)	Current health expenditure (CHE) as percentage of gross domestic product (GDP) (%)
China	0.001011955	4.158211
India	0.000523971	3.3362591
Pakistan	0.000652772	2.4296765
Bangladesh	0.000606792	2.2880766
Nigeria	0.000558256	3.2965329
Russia	0.000468014	4.971118
USA	0.00017749	16.197224
Indonesia	0.00021522	2.7908783

Ukraine	0.001117269	6.813876
Vietnam	0.000497974	5.0714607
Egypt	0.000426616	4.1531916
Germany	0.000422702	10.831174
Turkey	0.000442539	5.0205345
Iran	0.000350154	6.2915306
Japan	0.000196356	9.063006

Written Results and Interpretation (0.5 point)

We examined whether the share of GDP a country spends on healthcare (CHE as % of GDP, 2010) is related to its premature mortality rate (death per capita, 2010). ACross the 15 countires, CHE ranged from 2.29% (Bangladesh) to 16.20% (USA), with a mean of about 5.78%. Premature mortality rates ranged from 0.000176 (USA) to 0.001117 (Ukraine), with a mean of about 0.000511 deaths per capita. A Pearson correlation showed a weak-to-moderate negative relationship between the two variables (r=-0.36).

Interpretation

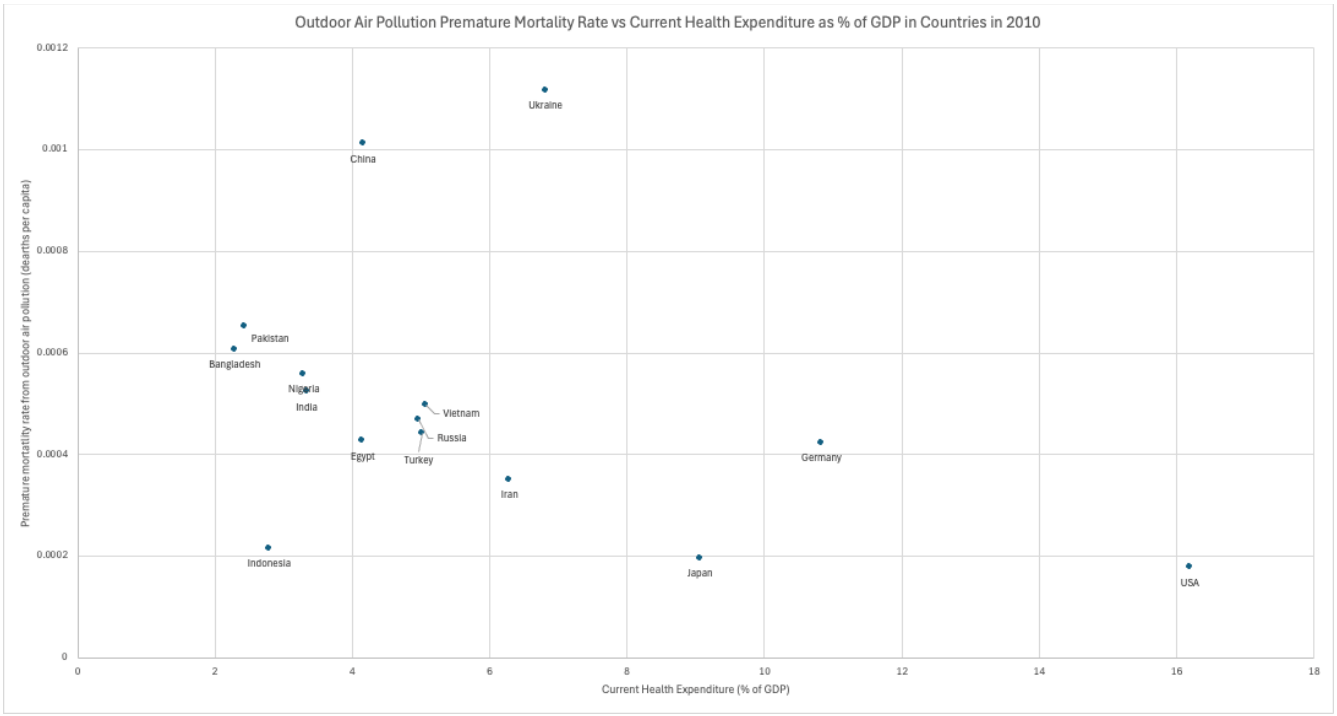
This data shows a tendency, not a clear relationship. The negative direction fits the intuition that more health spending leads to better outcomes, and the clearest examples are Japan (9.1% CHE, 0.000196 premature mortality rate), and the USA (16.2% CHE, 0.000177 premature mortality rate) which have some of the lowest mortality rates. But the pattern is inconsistent. Ukraine (6.8% CHE) has the highest mortality rate in the dataset, while Indonesia (2.8% CHE) has one of the lowest. Germany spends over 10% of GDP on healthcare, yet its mortality rate (0.000423) is similar to Egypt's (0.000427) at 4.2%.

Several factors may explain this weak relationship. First, spending efficiency matters. A percentage of GDP does not showl how this money is spent; spending on preventative care, primary care, and public health can lead to lower premature mortality more than spending on high-cost specialist care. Second, and most importantly, exposure is the dominant factor in deaths from outdoor air pollution. Healthcare can treat the illnesses that pollution causes, but higheer spending does not offset high exposure to ambient PM2.5 driven by coal use, vehicle emissions, and open burning. This may help explain why China and Ukraine, with moderate spending, have the highest rates. Finally, wealth is likely a confounding factor. Wealthier countries tend to spend a larger share of GDP on healthcare and also have stricter emissions standards and cleaner energy sources, so the observed negative correlation may reflect level of development rather than the effect of healthcare spending itself.

Conclusion

Because outdoor air pollution mortality is driven mainly by exposure, health expenditure as a percentage of GDP is only an indirect and weak predictor of it. The modest negative correlation we observed is likely explained largely by wealth and environmental regulation rather than by healthcare spending itself. Health spending might still be useful as a secondary variable, ideally alongside GDP per capita, and future analysis would be improved by including direct measures of exposure such as average PM2.5 concentration.

Figure (0.5 point)


Outdoor Air Pollution Premature Mortality Rate vs Current Health Expenditure as % of GDP in Countries in 2010
Premature mortality rate from outdoor air pollution (deaths per capita)
Current Health Expenditure (% of GDP)
0.0012
0.001
0.0008
0.0006
0.0004
0.0002
0
0
2
4
6
8
10
12
14
16
18
Ukraine
China
Pakistan
Bangladesh
Nigeria
India
Vietnam
Russia
Egypt
Turkey
Iran
Germany
Indonesia
Japan
USA